SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B.TECH (AI&DS)	**Date:**	

Code of Conduct

*I **VIVETHA B 192424170** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **VIVETHA B**

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

1. Semantic Representation in Customer Support Chatbots.
   A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

1. Analyze the Semantic Representations and Action-Object Relationships

The semantic representations identify the **action**, **object**, and **customer** involved in each query.

Query 1: "Activate international roaming for my number."

Semantic representation:

ACTIVATE(Roaming, Customer)

- **Action:** ACTIVATE
- **Object:** Roaming
- **Entity:** Customer

The customer wants to **activate international roaming**.

Query 2: "Deactivate caller tune service."

Semantic representation:

DEACTIVATE(CallerTune, Customer)

- **Action:** DEACTIVATE
- **Object:** CallerTune
- **Entity:** Customer

The customer wants to **deactivate the caller tune service**.

Query 3: "Check my data balance."

Semantic representation:

QUERY(DataBalance, Customer)

- **Action:** QUERY
- **Object:** DataBalance
- **Entity:** Customer

The customer wants to **check their available data balance**.

Query 4: "Enable 5G service."

Semantic representation:

ACTIVATE(5GService, Customer)

- **Action:** ACTIVATE
- **Object:** 5GService
- **Entity:** Customer

The customer wants to **activate/enable 5G service**.

Therefore, the semantic representations correctly capture the **action-object relationships** for all four queries.

2. Query Containing Semantic Interpretation Errors

From the chatbot logs:

Q1:

Actual: **Activate Roaming**

Predicted: **Activate Roaming**

Correct.

Q2:

Actual: **Deactivate Caller Tune**

Predicted: **Activate Caller Tune**

Incorrect.

The semantic representation is:

DEACTIVATE(CallerTune, Customer)

Therefore, the chatbot should identify the action as **DEACTIVATE**, not **ACTIVATE**.

This is the main semantic interpretation error.

Q3:

Actual: **Check Data Balance**

Predicted: **Query Data Balance**

Correct.

"Check" is represented by the action **QUERY**.

Q4:

Actual: **Enable 5G Service**

Predicted: **Activate 5G Service**

Correct.

"Enable" and "Activate" represent the same intended action in this context.

Therefore, **Q2 contains the semantic interpretation error**.

3. Effect of Meaning Representation on Chatbot Decision Accuracy

Meaning representation converts natural-language queries into a structured form that the chatbot can understand.

For example:

"Deactivate caller tune service."

is represented as:

DEACTIVATE(CallerTune, Customer)

This representation clearly identifies:

Action → DEACTIVATE

Object → CallerTune

The chatbot can then use this information to perform the correct operation.

If the semantic representation is incorrect, the chatbot may perform the wrong action.

For example:

DEACTIVATE(CallerTune, Customer)

being incorrectly interpreted as:

ACTIVATE(CallerTune, Customer)

could cause the system to activate a service that the customer actually wanted to deactivate.

Therefore, accurate semantic representation is essential for:

- **Intent detection**
- **Correct service execution**
- **Customer satisfaction**
- **Response generation**
- **Reducing incorrect actions**
- **Improving chatbot reliability**

4. Recommended Improvements

The telecom company can improve semantic understanding using the following methods.

1. Improve intent classification

Train the chatbot with more examples of similar commands such as:

Activate → Enable → Start

Deactivate → Disable → Stop → Cancel

This helps the system understand different expressions of the same intent.

2. Use context information

The chatbot should consider the complete sentence rather than individual words.

For example:

"Activate caller tune"

and

"Deactivate caller tune"

contain the same object but have completely different actions.

3. Use entity and action extraction

The system should separately identify:

Action → Activate/Deactivate/Query

Object → Roaming/CallerTune/DataBalance/5GService

Customer → User

This reduces confusion between actions and objects.

4. Handle synonyms

The system should recognize that:

Enable ≈ Activate

and:

Disable ≈ Deactivate

This is particularly important in customer conversations where users may use different words for the same operation.

5. Use contextual semantic models

Modern NLP models can be trained using customer-support conversations to understand the meaning of complete queries and their context.

6. Add confirmation for sensitive actions

For important service changes, the chatbot can confirm the customer's intention before executing the action.

For example:

"You want to deactivate Caller Tune. Shall I proceed?"

This prevents incorrect service operations.

2. First-Order Predicate Calculus for Smart Manufacturing.
   An Industry 4.0 manufacturing plant monitors machines using semantic rules.
   Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- Active(x) → Producing(x)
- Produces(x,y) ∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x)

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

1. Represent the Production Data Using First-Order Predicate Calculus

The machines and their statuses can be represented using predicates.

Given:

- M1 → Active
- M2 → Active
- M3 → Maintenance
- M4 → Active

The facts are:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

The given rules can be represented as:

∀x [Active(x) → Producing(x)]

This means that if a machine is active, it is producing.

∀x∀y [(Produces(x,y) ∧ Active(x)) → Available(y)]

This means that if an active machine produces a product, that product is available.

∀x [Maintenance(x) → ¬Producing(x)]

This means that a machine under maintenance is not producing.

Therefore, from the production data:

Producing(M1)

Producing(M2)

¬Producing(M3)

Producing(M4)

2. Analyze Which Products Are Currently Available

The rule is:

Produces(x,y) ∧ Active(x) → Available(y)

However, the question does not provide which products are produced by M1, M2, M3, and M4.

If we assume that the production relationships are:

Produces(M1, Gear)

Produces(M2, Wheel)

Produces(M3, Gear)

Produces(M4, Shaft)

Then:

For M1:

Produces(M1,Gear) ∧ Active(M1)

Therefore:

Available(Gear)

For M2:

Produces(M2,Wheel) ∧ Active(M2)

Therefore:

Available(Wheel)

For M3:

Produces(M3,Gear) ∧ Maintenance(M3)

M3 is not active, so it cannot establish:

Available(Gear)

For M4:

Produces(M4,Shaft) ∧ Active(M4)

Therefore:

Available(Shaft)

Hence, under this assumed production mapping:

Available(Gear)

Available(Wheel)

Available(Shaft)

3. Infer Whether Gear Production Is Affected by Maintenance

M3 is under maintenance:

Maintenance(M3)

Using the rule:

Maintenance(x) → ¬Producing(x)

we get:

¬Producing(M3)

If M3 produces Gear:

Produces(M3,Gear)

then M3 cannot currently produce Gear.

Therefore:

Gear production by M3 is affected by maintenance.

However, if another active machine such as M1 also produces Gear:

Produces(M1,Gear) ∧ Active(M1)

then:

Available(Gear)

So, maintenance of M3 affects **M3's Gear production**, but it does **not necessarily stop overall Gear availability** if another active machine produces Gear.

4. Effectiveness of Predicate Logic in Industrial Decision-Support Systems

First-Order Predicate Calculus is useful in industrial systems because it represents **machines, products, statuses, and relationships** in a structured logical form.

For example:

Active(M1) → Producing(M1)

allows the system to automatically infer that M1 is producing.

Similarly:

Maintenance(M3) → ¬Producing(M3)

allows the system to identify that M3 has stopped production.

Predicate logic is useful for:

- **Machine status monitoring**
- **Production planning**
- **Fault detection**
- **Maintenance management**
- **Product availability analysis**
- **Automated decision-making**
- **Rule-based reasoning**

It also provides explainable decisions because each conclusion can be traced back to a specific rule and fact.

However, predicate logic alone may not handle uncertain or noisy sensor data very well. In real Industry 4.0 systems, it can be combined with **machine learning, probabilistic models, and real-time sensor data**.

3. Word Sense Disambiguation in E-Commerce Search Engines.
   An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand

"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

1. Analyze the Context and Determine the Correct Word Sense

Query 1: "Apple accessories"

The clicked result is **"iPhone Charger."**

An iPhone is a product of the technology company Apple. Therefore, the intended meaning of **Apple** is:

Technology Brand

Correct sense:

Apple → Technology Brand

Query 2: "Mouse wireless"

The clicked result is **"Bluetooth Mouse."**

A Bluetooth mouse is a computer peripheral, so the intended meaning of **Mouse** is:

Computer Device

Correct sense:

Mouse → Computer Device

Query 3: "Java tutorial"

The clicked result is **"Coding Lessons."**

A tutorial related to coding indicates that Java refers to the programming language.

Correct sense:

Java → Programming Language

Query 4: "Python course"

The clicked result is **"Software Development Training."**

Software development training indicates that Python refers to the programming language.

Correct sense:

Python → Programming Language

2. Semantic Cues Used for Disambiguation

The search engine uses surrounding words and clicked-result information as semantic cues.

For **"Apple accessories"**, the word **"accessories"** suggests products related to a technology brand rather than fruit.

For **"Mouse wireless"**, the word **"wireless"** strongly suggests a computer device because wireless mice are common computer peripherals.

For **"Java tutorial"**, the word **"tutorial"** combined with the clicked result **"Coding Lessons"** indicates the programming language.

For **"Python course"**, the word **"course"** and the clicked result **"Software Development Training"** indicate the programming language.

Important semantic cues include:

- **Surrounding words**
- **Query context**
- **Clicked results**
- **Product categories**
- **User search behavior**
- **Domain-specific terminology**

3. Impact of Incorrect Sense Selection

Incorrect Word Sense Disambiguation can significantly reduce search engine performance.

For example, if **"Apple accessories"** is interpreted as fruit, the search engine might recommend:

Fruit baskets, fruit cutters, or apple-related food products

instead of:

iPhone chargers, AirPods, or phone accessories.

Similarly, interpreting **"Python course"** as a snake could produce irrelevant results about:

Snake handling or wildlife

instead of:

Python programming courses.

Incorrect sense selection can lead to:

- **Irrelevant search results**
- **Poor recommendations**
- **Lower click-through rates**
- **Reduced customer satisfaction**
- **Lower conversion rates**
- **Poor personalization**
- **Loss of customer trust**

Therefore, accurate WSD is important for improving both **search relevance and e-commerce sales**.

4. Industrial-Scale WSD Strategy

An industrial e-commerce search engine can use a combination of **NLP, machine learning, and user behavior analysis**.

1. Context-based analysis

Analyze surrounding query words to determine the intended meaning.

For example:

"Apple accessories"

contains **"accessories"**, which strongly indicates the technology brand.

2. Use contextual language models

Modern transformer-based language models can understand the context in which an ambiguous word appears.

For example:

"Python programming course"

and

"Python snake habitat"

have very different meanings even though both contain the word **Python**.

3. Use product and category information

The search engine can compare the query with product categories.

For example:

"Mouse wireless" → Computer Accessories

"Apple accessories" → Mobile/Technology Accessories

4. Use click and purchase history

User behavior provides strong evidence about the intended sense.

If users searching for **"Java tutorial"** frequently click coding courses, the system can learn that this query usually refers to the programming language.

5. Combine multiple signals

An industrial system can combine:

Query context + Product metadata + User history + Click behavior + Language model

to determine the most likely word sense.

6. Continuous learning

The system should continuously learn from new searches, clicks, purchases, and user feedback so that its WSD performance improves over time.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument

Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

1. How Syntactic Structures Contribute to Semantic Role Assignment

Syntax-driven semantic analysis uses the **grammatical structure of a sentence** to determine the role played by each entity.

For example:

"Doctor prescribed medicine to patient."

The structure is:

Subject → Verb → Object

- **Doctor** → Subject → **Agent**
- **prescribed** → Action
- **medicine** → Object → **Theme/Instrument**
- **patient** → Recipient

Similarly:

"Patient reported severe headache."

- **Patient** → Subject → **Experiencer**
- **reported** → Action
- **severe headache** → Object → **Symptom**

Another example:

"Nurse monitored patient continuously."

- **Nurse** → Subject → **Agent**
- **monitored** → Action
- **patient** → Object → **Patient/Theme**
- **continuously** → Manner

Thus, the syntactic relationship between the **subject, verb, and object** provides important information for assigning semantic roles.

2. Appropriateness of the Assigned Semantic Roles

The given roles are:

Doctor → Agent

This is **appropriate** because the doctor performs the action of prescribing.

Medicine → Instrument

This is **not the best role** for this sentence.

In:

"Doctor prescribed medicine to patient."

the medicine is the **Theme**, meaning the entity being prescribed or transferred to the patient. It is not really an instrument used to perform the action.

A better assignment is:

Medicine → Theme

Patient → Recipient

This is **appropriate** because the patient receives the prescribed medicine.

Headache → Symptom

This is **appropriate** because the headache represents a medical symptom reported by the patient.

Therefore, the main correction is:

Medicine → Theme, rather than **Instrument**.

3. Potential Errors from Incorrect Parsing

Incorrect syntactic parsing can cause serious problems in healthcare NLP systems.

For example, consider:

"Doctor prescribed medicine to patient."

If the system incorrectly identifies **patient** as the subject instead of the recipient, the semantic interpretation could become incorrect.

Incorrect parsing may cause:

1. Incorrect medical relationships

The system may incorrectly associate a medicine with the wrong patient.

2. Incorrect symptom identification

A symptom such as **headache** may be associated with the wrong person or medical event.

3. Incorrect treatment extraction

The system may fail to identify which doctor prescribed which medicine to which patient.

4. Incorrect clinical decision support

Wrong semantic relationships can lead to incorrect recommendations or alerts.

5. Patient safety risks

In healthcare applications, incorrect interpretation of medical information can potentially result in serious consequences.

Therefore, accurate parsing is especially important for medical NLP systems.

4. Methods to Improve Syntax-Driven Semantic Analysis

1. Use medical-specific NLP models

Train NLP models using medical reports and healthcare terminology so that the system understands medical language better.

2. Improve dependency parsing

Dependency parsing can identify relationships between words, such as:

Doctor → prescribed → medicine

and:

prescribed → to → patient

This helps assign semantic roles accurately.

3. Use semantic role labeling

Semantic Role Labeling can automatically identify:

Agent → Doctor

Action → Prescribed

Theme → Medicine

Recipient → Patient

Symptom → Headache

4. Use medical dictionaries and ontologies

Healthcare knowledge bases can help identify medical concepts such as:

Medicine, Disease, Symptom, Patient, Doctor, Treatment

5. Use contextual language models

Context-aware models can understand the meaning of medical terms based on the complete sentence rather than individual words.

6. Human validation for critical information

Important medical information should be validated by healthcare professionals, especially when the system is used for clinical decision-making.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

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